Asgmt 2:

Grade: Check Plus

Stock Price Distribution → Option Value Distribution

Stock Price X	f(x)
20	0.05
21	0.06
...	...
30	0.05
	1.00

(1) 1. Mean:

✓ Q1

$$\boxed{\mu = E(X) = \sum X f(x)} = 25.41$$

20 (0.05)
21 (0.06)
30 (0.05)
1.00

$\sum$

2. SD:

Probability Stock Price

$$\boxed{Var(X) = \sum f(x)(x-\mu)^2}$$

x	f(x)	x − μ	(x − μ)²	f(x)(x − μ)²
20	0.05	−5.45	29.2681	1.4634
21	0.06	−4.41	19.4481	1.1669
22	0.07	−3.41		
...	...			
30	0.05	4.59	21.0681	1.0534

Var(X) = 7.5019

3.

4. SD:

$$\boxed{\sigma = \sqrt{Var(X)}} = \sqrt{7.5019} \approx 2.74$$

Final: $Var(X) = \sum f(x)(x-\mu)^2 = 7.50$

$\sigma = \sqrt{7.50} = 2.74$

(2) a) Within 1 SD:

$(\mu - \sigma, \mu + \sigma) = (25.41 - 2.74, 25.41 + 2.74) = (22.67, 28.15)$

Prices within interval: ✓ Q2a

\$ 23 – 28

$\sum$ Probs:

$0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.10 = 0.67$

Prob within 1 SD = 0.67

b) Within 2 SD:

✓ Q2b

$(\mu - 2\sigma, \mu + 2\sigma) = (25.41 - 5.48, 25.41 + 5.48) = (19.93, 30.89)$

Prices:

\$ 20 – 30

Prob within 2 SD = 1.00

(3) Distribution of Option Value:

K = 25

V depends on X through: $V = \max(X - 25, 0)$

1) Table:

Stock Prices X: 20, 21, 22, ..., 30

Probs $f(x)$: 0.05, 0.06, 0.07, ..., 0.05

2) Option Payoffs:

K = 25

If $X \leq 25 \rightarrow V = 0$

$X = 26 \rightarrow V = 1$

$X = 27 \rightarrow V = 2$

...

$X = 30 \rightarrow V = 5$

3) $X = 21, 22, \ldots, 25 \rightarrow V = 0$

Payoff $V = 0$ happens when $X \leq 25$

$P(V=0) = 0.05 + 0.06 + \dots + 0.12 = 0.47$

Others:

$V = 1$ when $X = 26$, $P = 0.15$

2 $X = 27$, $P = 0.13$

3 $X = 28$, $P = 0.10$

4 $X = 29$, $P = 0.10$

5 $X = 30$, $P = 0.05$

Distribution of V: $V = \max(X - 25, 0)$

Option Value V	Prob. P(v)
0	0.47 ✓ Q3
1	0.15
2	0.13
3	0.10
4	0.10
5	0.05
	= 1.00

(4) Probability _Option _Positive Value:

✓ Q4

$V > 0 \rightarrow X > 25$

$P(V>0) = 1 - P(V=0) = 1 - 0.47 = 0.53$ OR 53%

(5) Mean & SD:

1) Expected Value (μ) E(V):

$$\boxed{E(V) = \sum V \cdot P(V=v)}$$

$E(V) = 0(0.47) + 1(0.15) + 2(0.13) + 3(0.10) + 4(0.10) + 5(0.05)$

$= 1.36$

✓ Q5

$\mu_V = 1.36$

2) $$\boxed{E(V^2) = \sum v^2 \cdot P(V=v)}$$

$E(V^2) = 0^2(0.47) + 1^2(0.15) + \dots + 5^2(0.05) = 4.42$

3) Variance (V) $= E(V^2) - (E(V))^2 = 4.42 - 1.36^2 = 2.5704$

$\sigma(V) = \sqrt{Var} = \sqrt{2.5704} \approx 1.603$